

AD-202 Manual

Audio Damage, Inc.

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System Requirements

The following table summarizes the operating system requirements and formats provided for AD-202. AD-202 is a 64-bit plugin.

Operating System	Minimum Version	Formats
macOS	10.13	AAX, AudioUnit, CLAP, VST3; Intel and Apple Silicon
Windows	10.0 version 1607	AAX, CLAP, VST3
Ubuntu	20.0	CLAP, LV2, VST3
iOS (separate purchase)	iOS 16	AUv3

Demonstration Version

We encourage you to download and try the demonstration version of AD-202 before purchasing it. The demo version of AD-202 is the same as the regular version, but has the following limitations:

- Presets cannot be saved, nor can parameter values or other settings. This includes the information usually stored by your host DAW. If you save a DAW session with an instance of the demo version of AD-202, the plugin will revert to its default state when you reload the session.
- AD-202 will cease to emit audio altogether 20 minutes after you add it to your DAW session. You can remove it and add it again, but it will revert to its default state.

If you purchase AD-202 after using the demonstration version, simply run the installer provided to you after your purchase to replace the demo version with the full version.

Introduction

AD-202 is a monophonic synthesizer with one oscillator, one filter, one envelope, and one LFO.

That sentence is either a warning or an invitation, depending on your temperament. If you're used to instruments with six oscillators and a modulation matrix, a single-oscillator monosynth can look like a toy. It isn't. Some of the most recognizable sounds in the last forty years of recorded music came out of instruments built exactly this way, and there is a reason for that: when there is only one of everything, every control matters, and you can hear what each one does. You stop configuring and start playing.

When we conceived of this plugin, the instrument we had in mind was the MC-202, and the SH-101 whose voice circuit it borrowed. Those two share a design that is close to ideal for its size. A single oscillator produces a sawtooth and a pulse whose width you can narrow by hand or by modulation. A sub-oscillator, formed by a simple divider hung off the main oscillator and always perfectly in tune, makes these instruments punch far above their weight in the bass. A noise source rounds out a four-input mixer that deliberately clips on its way into a four-pole ladder lowpass filter. One envelope opens the filter and shapes the amplitude at the same time. One LFO wobbles whatever you point it at. That's the whole instrument, and it is enough.

We built AD-202's voice from both hands-on experience and service-manual research, and kept the parts of the original's behavior that a more "correct" design would have quietly improved away. The pulse width only narrows, from square toward a sliver, because that is the only direction the original circuit could go. Pulse-width modulation from the LFO always uses the triangle wave no matter what the LFO shape switch is set to, because that is how the circuit was wired. The filter gets *louder* as you add resonance rather than thinner, which is a side effect of how the hardware's output stage was arranged and is a large part of why these instruments sound aggressive rather than polite when you open them up. The mixer clips. If any of these design decisions were accidental, history has shown that none of them are defects.

Once we'd nailed the voice, we went further to build a software plugin with modern conveniences not available to hardware builders of last century. The original was a monosynth attached to a sequencer, and the sequencer was most of the point of it in 1983. We spent a while deciding whether a plugin needs one, given that you are already sitting in front of a DAW that is a far more capable tool for constructing patterns than any Microcomposer. It turns out that it does, for a reason that has nothing to do with capability: a small sequencer you can reach into while it is playing finds parts that a piano roll editor never suggests. Rotate a pattern one step to the left and the notes stay exactly where they were while the emphasis lands somewhere else entirely, and you have a part you would not have written down.

AD-202 will also run as an eight-voice polyphonic instrument when you want chords, with a stereo spread control to place those voices across the field. An Age control gives each voice its own permanent set of small mis-calibrations, the way an old instrument's voice circuits drift apart from one another over the decades. This detuning is pleasant in mono and something close

to a chorus in poly. Three separate saturation stages let you dirty the signal before the filter, inside the filter, or after the amplifier, each with its own character. There is a tilt EQ at the end to place the result in a mix. The LFO will lock to your host's tempo. Velocity and aftertouch, neither of which the original had any notion of, route to the places where they do the most good. Finally, the whole thing speaks MPE and MTS-ESP for those who want per-note expression or tunings other than the twelve we usually agreed on.

The panel reproduces the MC-202's, down to the sampled colors, the lever switches, and the chicklet keys. Five finishes are included. Four of them are colors. The fifth is not, quite, and if you recognize the name we don't need to explain it to you.

This manual gives an overview of AD-202 followed by detailed descriptions of all of its controls. You are welcome to install it, flip through the presets, and start pulling faders until something sounds good. With an instrument this size that is a genuinely efficient way to learn it, and we'd be a little disappointed if you didn't. But there is more here than the front panel admits, particularly in the MODS panel, and it's worth coming back to read once you've had a chance to play. We hope you enjoy using it as much as we enjoyed building it.

New in Version 1.0.5

- A sequencer. Click SEQ and a bassline step sequencer opens in a band below the keyboard: eight patterns of up to 64 steps, each step accented, slid into the next, or pushed by octaves either way; follow actions and repeat counts chain the patterns into an arrangement; swing, and a row of tools for randomizing, varying, copying, clearing and rotating. A realtime mutation panel alters the pattern on its way out without altering what you wrote. The standalone iOS version can utilize Ableton Link to stay in sync with other tools in your studio or onstage.
- On iOS the panel scrolls, so the keyboard strip and the sequencer are reachable on a small screen, and the standalone keeps playing when you send it to the background.

New in Version 1.0.1

- Sweeping the tilt EQ through its center no longer clicks. The control is smoothed now, so fast moves and automation no longer zipper.
- Every trigger mode is last-note priority. Gate and LFO previously sounded the lowest key held; all three settings now sound the most recently pressed key and differ only in when the envelope fires.
- Touch or click a control and the preset display shows that control's name and value while you hold it.
- A new preference, Jump To Click, for those who would rather a fader jumped to the mouse than dragged from where it sits. It is off by default.
- Latch is now a parameter, so automation and control surfaces can reach it. Its state is deliberately not stored in the preset.

Installation

AD-202 uses our custom plugin manager application for installation. Launch it as usual on your operating system of choice and you'll be presented with a window like this:



Near the top of the window, beneath the name of the plugin, you'll see the version number of the software carried by the installer. This is distinct from the version of the plugin manager itself, which is shown in the upper right and usually not of much interest. (Note that the version numbers in this screenshot are for demonstration purposes only and may not reflect the actual version numbers of the software at the time you read this manual.)

Under the heading FORMATS are large buttons corresponding to the plugin formats which can be installed: AAX, AU, CLAP, LV2 and/or VST3, depending on the operating system. If the plugin is already present on your system in one or more formats (i.e. if you're upgrading from a previous version), the corresponding button is drawn in blue. When possible, the version number of the existing plugin is also shown. If the installer contains a newer version of the plugin than the one(s) already installed, the button is drawn in orange to alert you to the fact that an update is available. If a format is not present on your system, its button is drawn in grey.

Click a button to select the format for installation. A yellow outline appears around the button to indicate that its format will be installed. In the above screenshot, AAX and AudioUnit are not installed, VST3 is installed but older than the version contained in the installer, CLAP is installed and up to date, and the AudioUnit is selected for installation. Clicking a button a second time removes the yellow outline, and the corresponding format will not be installed. Clicking the SELECT ALL button selects all available formats for installation.

No changes to your system's storage device take place until you click the INSTALL button near the lower-right corner of the window. Click that button and you'll receive visual confirmation that the format(s) you've selected have been installed. (Yes, it happens quickly. Our products are not encumbered by any DRM or anti-piracy baggage and hence install more quickly than many others.) On Windows and Linux, if you hold down the Shift key on your keyboard, the INSTALL button's label switches to UNINSTALL, and clicking it will remove the selected formats from your system¹. Once you've installed and/or removed the formats you need, simply close the application in the usual manner for your operating system. You're done. There is no license code or other authorization necessary; we'd rather assume we can trust you than burden you with an onerous DRM system.

You'll find some handy buttons under the EXTRAS heading, all of which are pretty self-explanatory:

- EULA: presents the End-User License Agreement for our products. By clicking the INSTALL button you're implicitly agreeing to these terms, but we expect that you'll find them reasonable should you take the time to read them.
- MANUAL: opens the current version of this user manual, in PDF form, in your web browser.
- WEBSITE: opens the product's web page in your browser.

¹ Blame Apple, not us, for the lack of this feature on macOS. On macOS just manually delete the plugin(s) from your plugin folder(s).

- **SUPPORT:** displays information for contacting us, either via our Discord presence or through email.
- **SETTINGS:** produces a panel in which you can choose the installation locations for each plugin format. Note that these locations are nominally stipulated by the publishers of the formats and changing them might cause problems for some hosts.
- **CHANGES:** displays a summary of what we've added, changed or improved since the last version of the product. This button isn't present on the first release of the product.

Operation

AD-202 is a monophonic (and, when you want it to be, polyphonic) subtractive synthesizer built around a single oscillator, a four-pole ladder lowpass filter, one shared envelope, and one LFO. This small feature set is the reason the instrument is easy to play: there is exactly one of everything, and every control does one obvious thing.



The main panel presents the synthesis controls in the six sections printed across the top half of the window: LFO, VCO, SOURCE MIXER, VCF, VCA, and ENV. The lower half of the panel holds the performance controls (tuning, portamento, trigger behavior, velocity response, and output level) along with the preset display and a couple of status indicators. Below the panel is a keyboard strip you can play with the mouse or your finger on touchscreen devices.

Two additional panels slide over the main window. MODS holds a set of secondary sound parameters that are saved with the preset: voice mode, the saturation stages, and the aftertouch routings. SETTINGS holds global preferences that are not saved with the preset. A third button, SEQ, opens a sequencer in a band below the keyboard rather than over the panel, making the window taller. All three are described below.

VCO

The oscillator section produces a sawtooth and a variable-width pulse from a single master oscillator, plus a sub-oscillator derived from that oscillator by a flip-flop divider chain. The saw and pulse are anti-aliased so they stay clean at the top of the keyboard.

RANGE is the octave switch, setting the footage of the oscillator: 16', 8', 4', or 2'. The default is 8', which puts middle C where you expect it to sound.

PULSE WIDTH sets the width of the pulse wave, or the depth of its modulation, depending on where the switch beside it is set. The control only narrows the wave: at minimum the pulse is a 50% square, and turning it up narrows the wave toward roughly 2%. This matches the behavior of the original hardware, where the width control could only take the wave away from square. Since a 50% square has no even harmonics, the fader's effect is dramatic near the bottom of its travel, changing the sound from hollow to nasal almost immediately, and then progressively thinner as you keep moving the slider upwards.

The three-position switch beside the Pulse Width fader chooses what controls the width:

- MAN: the fader sets the width directly, and it stays there.
- LFO: the LFO sweeps the width, with the fader setting the depth of the sweep. Note that the LFO source here is *always the triangle wave*, regardless of what the SHAPE switch in the LFO section is set to. This is how the original circuit was wired, and it is what makes the classic PWM string pad work no matter what else the LFO is doing.
- ENV: the envelope sweeps the width, with the fader setting the depth. Higher envelope output means a narrower pulse, so the wave thins out during the attack and fills back in during the release.

MOD sets the depth of LFO modulation of the oscillator's pitch: vibrato, in other words. At minimum there is none. At maximum the LFO swings the pitch across a full octave, which is well past vibrato and into siren territory. Unlike the pulse-width modulation, this one uses whichever waveform the LFO SHAPE switch has selected, so with the square wave you get a trill and with the random waveform you get a sample-and-hold pitch sequence.

Source Mixer

The Source Mixer combines the four sound sources and feeds them to the filter. The four faders are, from left to right, pulse, sawtooth, sub-oscillator, and noise. The first two are marked with waveform glyphs; the second two are marked SUB OSC and NOISE.

There is no gain staging to worry about beyond your own taste, but note that the mixer soft-clips on its way into the filter. Running all four faders up is not a mistake; it is a tone color, and it is where a lot of the instrument's grit comes from. If you want more of that grit, the PRE-FILTER SAT control in the MODS panel pushes the same clipper harder.

The pulse, sawtooth, and noise faders operate as you'd expect, setting the levels of the corresponding signal. The SUB OSC sets the level of the sub-oscillator, and the three-position switch beside it chooses which of three sub waveforms you get:

- -1 OCT square: a square wave one octave below the oscillator.
- -2 OCT square: a square wave two octaves below.
- -2 OCT pulse: a narrow pulse two octaves below, produced by combining the two divider stages. This gives a 75/25 duty cycle, which is thinner and buzzier than the square and sits better under a bass line without turning it to mud.

The sub is generated by a divider running off the master oscillator, so it is always exactly in tune with it. It is gently filtered above 6kHz and DC-blocked.

NOISE sets the level of the white noise generator. The noise is rolled off above 16kHz, which takes the hiss off the very top and matches the character of the original. Noise is useful in small amounts as a texture inside a filtered patch, and in large amounts as a source in its own right for wind and percussion sounds.

VCF

The filter is a four-pole (24dB/octave) resonant ladder lowpass. Five faders control it.

FREQ sets the cutoff frequency, from 10Hz to 20kHz. The taper is skewed so that the middle of the fader's travel lands around 1kHz, spreading the the useful range across more of the fader range. This slider sets the base value of the filter frequency; the modulation sources described below add to this setting.

RES sets the resonance, from none at the bottom to self-oscillation at the top of the travel. As on the original, the filter's output level rises with resonance rather than falling, which is a consequence of how the hardware's phase splitter was arranged. The practical effect is that turning up the resonance makes the patch louder and more aggressive rather than thinner, so you may find yourself pulling the Volume knob back as you go.

ENV sets how much the envelope opens the filter. The modulation is positive-only and spans up to nine octaves at full, which is the entire range of the filter. It may be the single most expressive control in the instrument: it is what turns a static tone into a plucked, swept, or wobbling one. Note that the envelope driving it is the same envelope driving the VCA, so the filter sweep and the amplitude contour are always locked together.

MOD sets how much the LFO modulates the cutoff. This one is bipolar around the current cutoff setting, spanning up to plus or minus four octaves at full. With the triangle waveform this gives you the classic filter wobble; with the square it gives you a two-step alternation; with the random and noise waveforms it gives you burbling and hiss respectively.

KYBD sets the keyboard tracking of the cutoff, that is, how much the note you play moves the filter along with it. At 100% (the default) the cutoff tracks the keyboard one-for-one, so the timbre stays consistent as you move up and down the keyboard. At 0% the cutoff stays where you put it regardless of pitch, which means high notes get progressively duller and low notes progressively brighter. Tracking is referenced to middle C, so that note sounds the same at any Kybd setting.

The filter also has a drive control, but it lives in the MODS panel with the other saturation stages. See PRE-FILTER SAT and FILTER DRIVE below.

VCA

The VCA section is a single two-position switch which selects what controls the amplifier:

- ENV (upper position, marked with an envelope glyph) — the shared envelope controls the amplitude, so the sound follows the ADSR contour.
- GATE (lower position, marked with a gate glyph): the amplifier follows the key gate directly, snapping on when a key goes down and off when it comes up. There is a two-millisecond smoothing lag on the gate so it doesn't click.

Gate mode does not disable the envelope; it just takes the envelope out of the amplitude path. The envelope keeps running and keeps modulating the filter, so a common trick is to set Gate mode and use the envelope purely as a filter contour, giving you an organ-like amplitude with a moving timbre.

ENV

One envelope generator serves the filter, the amplifier, and optionally the pulse width. It is a conventional ADSR with the segment curvature of an analog envelope rather than the straight lines of a digital one.

- A (Attack) sets the time from key-down to full level, from 1.5ms to 4 seconds.
- D (Decay) sets the time from full level down to the sustain level, from 2ms to 10 seconds.
- S (Sustain) sets the level held while the key is down, from silence to full.
- R (Release) sets the time from key-up back to silence, from 2ms to 10 seconds.

The Attack, Decay, and Release faders are skewed so that the short times occupy a generous portion of the travel, since that is where percussive sounds live. Retriggering the envelope re-enters the attack segment *from the current level* rather than from zero, which is what an analog envelope does and what makes fast repeated notes sound continuous instead of chopped. Whether a given note retriggers the envelope at all depends on the TRIGGER switch; see below.

LFO

The LFO runs continuously, whether or not a note is held, and the red LED at the top of the section blinks along with it.

RATE sets the LFO speed, from 0.1Hz to 30Hz. If the SYNC switch below is set to ON, the fader instead selects a metrical division of the host tempo, from two whole bars down to a thirty-second note, with dotted and triplet values in between.

DELAY sets a fade-in time for the LFO, from zero to 4 seconds. With Delay up, the LFO's depth ramps up from nothing over the delay time after each note starts, rather than being applied immediately. This is the standard way of getting delayed vibrato, where a note starts steady and then begins to waver. The delay affects the LFO's *depth* only, not its phase, so the LFO keeps running underneath and the trigger behavior described below is unaffected. The ramp restarts on each new note attack.

SHAPE selects the LFO waveform from four positions, top to bottom:

- Triangle: a symmetrical rise and fall. The default, and what you want for vibrato and filter wobble.
- Square: an instant alternation between two values.
- Stepped random: a new random value at the start of each LFO cycle, quantized to a coarse staircase the way the original's six-bit converter was. The stepping is audible and is part of the character.
- Noise: a continuously changing random value updated at control rate, which is much faster and rougher than the stepped random.

SYNC locks the LFO rate to the host tempo, changing the Rate fader from a frequency control to a note-division control as described above.

RETRIG restarts the LFO's cycle from the beginning at each new note. With Retrigger off the LFO free-runs, so a given note starts wherever the LFO happened to be, and repeated notes get slightly different modulation each time. With it on, every note gets identical modulation, which is what you want when the LFO is doing something rhythmic.

Performance Controls

These sit across the lower half of the panel.



TUNE is the master tuning control, plus or minus 50 cents, marked with flat and sharp symbols at the ends of its travel. This tunes the whole instrument, not individual voices.

PORTAMENTO sets the glide time between notes, from zero to five seconds, marked SHORT and LONG. The glide is exponential in pitch, meaning it moves at a constant rate in octaves per second, so a glide across an octave takes the same time no matter where on the keyboard it happens.

MODE, the switch beside the Portamento knob, sets when the glide applies:

- ON: every note glides from the previous one.
- AUTO: notes glide only when played legato, that is, when a new key goes down while the previous one is still held. Staccato notes jump straight to pitch. This is the default and it is the most musically useful of the three, since it lets you choose note by note whether to glide.
- OFF: no glide.

TRIGGER sets the retrigger behavior. In all three settings the most recently pressed key is the one that sounds; what changes is when the envelope fires:

- GATE: single triggering. The envelope fires only when you go from no keys held to one, so playing legato moves the pitch without restarting the envelope.
- GATE + TRIG: multiple triggering. Every new key restarts the envelope. This is the default, and it is the behavior most people expect from a keyboard.
- LFO: the envelope retriggers on every cycle of the LFO for as long as a key is held. This turns the LFO into a repeat clock and the instrument into a self-playing arpeggiator of one note. Set a short envelope, turn the Rate up, and hold a key.

VEL-VCA sets how much MIDI velocity controls loudness. At minimum, every note plays at full level regardless of how hard you hit it, which is how the original behaved (it had no velocity sensing at all). Turning it up makes soft notes quieter.

VEL-VCF sets how much velocity controls the filter cutoff. As with Vel-VCA, the effect is subtractive: at higher settings, soft notes pull the cutoff down, by as much as four octaves at maximum. Notes played at full velocity always sound at the cutoff you dialed in, so this control never brightens a patch beyond where you set it; it only darkens the quiet notes.

VOLUME is the output level, running from silence to +6dB, marked MIN and MAX. It sits at the very end of the signal path, after the distortion and tilt EQ stages in the MODS panel.

Keyboard Strip

Below the panel is a 32-key keyboard running from F to C, the same span as the original's, which you can play with the mouse or your finger on a touchscreen. It is a convenience for auditioning patches, not a performance instrument: the on-screen keys have no velocity sensitivity and always play at a fixed medium velocity. If you want to hear what the Vel-VCA and Vel-VCF controls do, play the plugin from a MIDI keyboard.

OCT + and OCT - shift the keyboard up and down by one octave at a time, up to two octaves in either direction from the default position.

LATCH holds the keys you press until you press them again, so you can leave a note or a chord sounding while you work on the patch with both hands free. Turning Latch off releases everything currently held. Latch applies to incoming MIDI as well as to the on-screen keys. It is a parameter, so your host can automate it and a control surface can reach it, but it is deliberately not stored in the preset: a patch that loaded with a note already held would be a nuisance rather than a convenience.

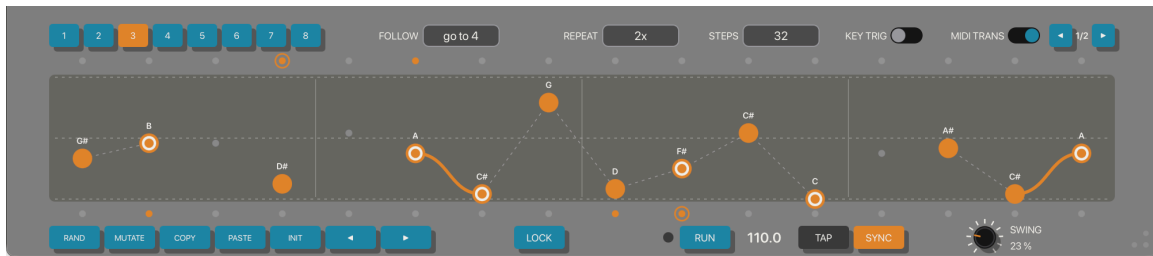
Two indicators sit above the keyboard, beside the preset display. MIDI lights when the plugin receives MIDI input, which is the first thing to check when a patch is silent. MTS lights when an MTS-ESP tuning master is running, in which case AD-202 takes its tuning from that master rather than from standard equal temperament, and notes that the master leaves unmapped will not sound.

The preset display doubles as a readout while you work. Touch or click any control and the display shows that control's name and its current value for as long as you hold it, which saves guessing at a fader's exact setting or reading a knob's position off the panel.

Sequencer

The MC-202 was a synthesizer bolted to a sequencer, and in 1983 the sequencer was most of the reason to own one. Nowadays a step sequencer built into a plugin is somewhat redundant, but it does invite a particular approach to building patterns that may not be present in your DAW of choice. We deliberately didn't the original's Microcomposer step entry, which was a marvel of industrial design and is no fun whatsoever with a mouse. Instead we've given AD-202 a more contemporary sequencer, suitable for both mouse-based and touchscreen interaction. Its design is directly inspired by Bram Bos' excellent Rozeta Bassline; we metaphorically tip our hat to him.

Clicking SEQ opens it in a band below the keyboard. The window keeps its width and grows taller to make room; clicking SEQ again closes it and the window shrinks back. The band stays as you left it if you close and reopen the plugin's window, but a new instance always starts with it closed.



The Pattern

The large field is the pattern itself. Each column is one sixteenth note, and the height of a note's dot is its pitch, across two octaves and the C on top. A thin, vertical line indicates which step is currently playing.

Click an empty column at the height you want and a note appears there. Click a note again, without dragging, and it cycles through its three states: on, accented, and off. Drag a note up or down to change its pitch; the note names change as you go, so you needn't count rows.

Two narrow rows of dots sit above and below the field. Clicking one of them shifts that step by an octave, clicking a second time by two octaves, and a third time puts it back where it started. As you might surmise, the upper row shifts notes higher in pitch while the lower row shifts them down. The two exclude each other; if you've got a downward shift on a step and you click its upper dot, the downward shift turns off.

Adjacent notes are connected with a dotted line. Click it and the first note slides into the second: the pair is played legato, so the instrument glides from one pitch to the next rather than retriggering. This produces the effect that defined an entire genre of electronic music, and with the PORTAMENTO MODE switch set to AUTO it does exactly what you want. A plain note lasts about three-fifths of its step; a note that slides holds on until the next one arrives.

Accented notes reach the instrument at full velocity while unaccented ones are a little softer. This matters more than it sounds like it should, because it means VEL-VCA and VEL-VCF on the main panel decide what an accent actually does. With both of those at minimum an accent changes nothing at all. Turn Vel-VCF up and the accented notes open the filter, which is probably what you want to hear.

Patterns, Length and Follow

The eight numbered tabs choose which of the eight patterns you are editing. Switching while the sequencer runs is handled politely: the new pattern takes over when the current one reaches its end, rather than cutting in partway.

STEPS sets the pattern's length in sixteenths. Every length from 1 to 16 is available, and then 24, 32, 48 and 64. The field shows sixteen steps at a time, so for anything longer the PAGE arrows move you a bar at a time. Yes, this gets jumpy while the sequencer is running so you'll probably want to stop playback while you're editing patterns that occupy more than 16 steps.

FOLLOW decides what happens when the pattern reaches its end. MANUAL simply loops it. NEXT and PREVIOUS switch to the neighboring pattern, RANDOM jumps to any of the eight, and GO TO 1 through GO TO 8 change to the specific pattern. REPEAT sets how many times the pattern plays before the follow action fires, from once to four times.

Between them these two build an arrangement out of short patterns without involving your DAW at all: e.g. play pattern 1 four times, then pattern 2 twice, then back again.

Playing From the Keyboard

Two switches decide what incoming notes do while the sequencer runs. They apply to the on-screen keys and to MIDI from outside alike.

MIDI TRANPOSE, which is on by default, turns the keyboard into a transpose control rather than a way of playing notes. C3 is home; play a D and the whole pattern moves up a tone.

KEY TRIG makes the pattern wait for you. With it on, the sequencer stays silent until you press a key, then plays the pattern from its first step, transposed to that key, and stops when you let go. This is the usual way to play a bassline part by hand, and LATCH works with it if you would rather not hold the key down.

With both switches off, the keyboard plays the instrument in the ordinary way and the pattern runs underneath at the pitches you wrote.

The Tools

A row of buttons runs along the bottom of the panel for executing various editing operations on the patterns.

COPY and PASTE let you move a whole pattern from one tab to another. INIT empties the current one. The two arrows rotate the pattern a step to the left or right, which is very useful if you're like the author of this manual and you find that you somehow tend to build patterns with the "one" in the the wrong place.

RANDOM opens a small panel of its own, allowing you to roll dice (metaphorically speaking) to make new patterns. KEY and SCALE decide what notes the dice are allowed to draw from. You'll find the familiar modes, both pentatonics, the blues scale, or Chromatic for no constraint at all. STEPS sets how long the result will be. GO writes a new pattern. VARY leaves the notes and rests where they are and re-rolls only the octaves, accents and slides.

MUTATE opens a panel of four knobs under the heading REALTIME MUTATION. These do not touch the stored pattern: they manipulate the notes on their way out, so turning the knobs back down returns the pattern to precisely what you wrote. ORDER is the chance that a note borrows its pitch from another note in the pattern. ACCENT and SLIDE are the chances that a note's accent or slide flips from on to off or vice-versa. SKIP is the chance that a step is dropped and the following one played in its place, which walks the pattern out of step with the bar and is much the most destructive of the four. All of them sit at zero until you move them, and small amounts go a long way.

The Transport Strip

The controls across the bottom right of the band control the sequencer's clock.

RUN starts and stops it. The dot to the left of RUN flashes on the beat while the sequencer is rolling, orange on the first beat of the bar and red on the other three, so you can see where you are without counting.

SYNC chooses the clock source. With it off, the sequencer keeps its own time at the tempo in the box beside it. With it on, the sequencer follows your host: the DAW's transport starts and stops it, the DAW's position places it within the bar, and the DAW's tempo takes over. The box then shows the host's tempo for reference rather than accepting a new one.

The tempo box holds the sequencer's own tempo, anywhere from 20 to 300 BPM. Click it and type a figure, or hit TAP four or more times at the tempo you have in mind. Both are inactive under host sync.

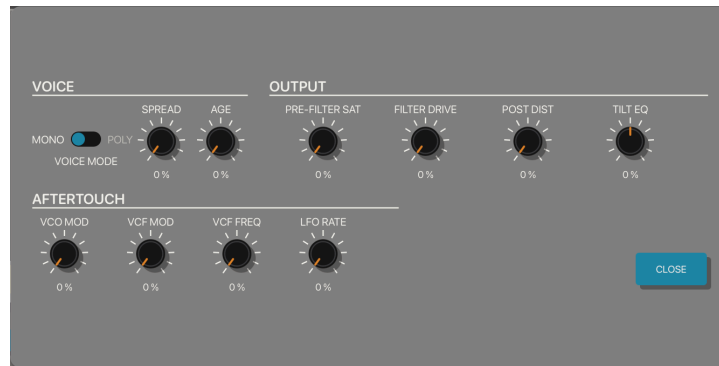
SWING delays every second sixteenth note, from no delay at all at zero up to half a step at maximum. Somewhere around two-thirds of the way up you will find a convincing triplet shuffle. Beyond that it goes places the original never could, which may or may not be what you are after.

LOCK is the most useful control on the strip and the least self-explanatory. The sequence, its settings and the sequencer's parameters are all saved in the preset, which is what you want when you save a patch and the riff that goes with it. It is emphatically not what you want when you are auditioning presets to find a sound for a riff you have just written, because every preset you load arrives with its own sequence and yours is gone. Turn LOCK on and the sequencer stops paying attention to preset loads altogether: the patterns, the tempo, the swing and everything else are held in place while you browse sounds underneath them. Turn it off again and the next preset you load takes over as usual. Note that presets saved before version 1.0.5 carry no sequence at all, so loading one of those with Lock off empties the sequencer.

LINK appears only in the standalone iOS version of AD-202. It joins an Ableton Link session, so that AD-202 and every other Link-aware application on the network share a tempo and a bar line. The small figure in the corner of the button is the number of other peers in the session. While Link is running the tempo box shows the session's tempo and typing into it affects the whole session. SYNC greys out and has no say when Link is active. Start/Stop Sync and the Quantum setting live in the Settings panel, described below.

MODS Panel

Clicking the MODS button slides in a panel of secondary sound parameters. Everything here is a real parameter and is saved with the preset, unlike the contents of the Settings panel. Click CLOSE to dismiss it.



Voice

VOICE MODE switches between MONO and POLY. In Mono the instrument behaves like the hardware it is modeled on: one voice, with the keyboard priority and trigger rules described above under TRIGGER. In Poly you get eight voices, with each incoming note allocated to the oldest free voice and, when all eight are busy, stealing the oldest sounding one. Portamento in Auto mode applies in Poly only when a voice is reused while still sounding.

SPREAD pans the voices apart in the stereo field, alternating left and right of center. It has no effect in Mono, where the output is always centered. At minimum all voices are centered; at maximum they are spread to the edges. The panning is equal-power, so the overall level stays put as you open the spread.

AGE applies a fixed random detuning and mis-calibration to each voice, in the manner of an old instrument whose voice circuits have drifted apart. Each voice gets its own permanent set of offsets, applied to pitch (up to about 20 cents), pulse width, filter cutoff (up to a quarter octave), LFO rate, and the four mixer levels. In Mono this reads as a slightly-off instrument; in Poly it reads as a chorus, since the voices are now genuinely different from each other. Small amounts do a lot.

Output

These four controls are the saturation and tone-shaping stages, arranged in signal order.

PRE-FILTER SAT drives the source mixer harder into its soft clipper, ahead of the filter. At minimum the clipper is at its stock level and only clips when you have several mixer faders up. Turning it up pushes the mix into the clipper by as much as a factor of eight, which thickens the sound and adds harmonics *before* the filter gets to remove them. This is the most musical of the three distortion stages because the filter tames whatever it produces.

FILTER DRIVE adds saturation inside the filter's feedback path. This is a different flavor from Pre-Filter Sat: rather than dirtying the input, it changes how the filter itself behaves under load, softening the resonance and giving the whole filter a rounder, more compressed character. The drive costs level as you turn it up, so a makeup gain rides along behind it and keeps the loudness roughly constant.

POST DIST is a distortion stage after the VCA, at the very end of the voice path. Because it is after the amplifier, it distorts the envelope contour as well as the waveform, flattening the dynamics and making short notes sound louder than they are. It is the bluntest of the three, and it is the one to reach for when you want the patch to sound broken. It too has a makeup gain, so turning it up does not simply make things louder.

TILT EQ is a bipolar tilt equalizer applied after the distortion. Turning it left removes highs and adds lows; turning it right does the opposite. Straight up is flat. It is a fast way to place a patch in a mix without reaching for a separate EQ.

Aftertouch

The four knobs in this section route MIDI channel pressure (or, in MPE mode, per-note pressure) to four destinations. All four are additive: they add to whatever the corresponding panel control is already doing, so a patch with no vibrato at rest can have vibrato on demand.

VCO MOD adds LFO-to-pitch depth with pressure, which is the standard way to get vibrato under your fingers rather than permanently on.

VCF MOD adds LFO-to-filter-cutoff depth with pressure.

VCF FREQ opens the filter with pressure, by as much as four octaves at full. This is the most immediately expressive of the four: lean into a note and it brightens.

LFO RATE speeds up the LFO with pressure, by as much as a factor of four. Combined with VCO MOD, this gives a vibrato that gets both deeper and faster as you press, which is how a string player actually plays.

If your controller sends stepped or noisy pressure, the Pressure Smoother in the Settings panel will smooth it; see below.

Settings Panel

Clicking SETTINGS opens the global preferences. Unlike the MODS panel, nothing here is saved with the preset. Changes apply to the current instance immediately; click SAVE to make them the defaults for new instances. Closing the panel without saving leaves the changes in effect for this instance only.

SHOW TOOLTIPS turns the hover tooltips on and off.

PANEL COLOR selects the color of the panel. We started with the colors of the cases of the MC-202 and SH-101, added a contemporary black and another derived from nostalgia for a consensual hallucination.

JUMP TO CLICK changes how the faders respond. With it off, which is the default, clicking anywhere on a fader picks it up where it already sits and drags from there, so a click never moves the sound. With it on, the fader jumps to wherever you clicked. Neither is right for everyone: the first protects a patch you like, the second is quicker when you know where you are going.

MPE MODE enables MPE, in which each note arrives on its own MIDI channel and carries its own pitch bend and pressure. With MPE off, the plugin runs in the conventional single-channel mode where bend and pressure apply to everything sounding. Turn this on only if your controller actually sends MPE; turning it on otherwise will make bends behave strangely.

PITCH BEND OVERRIDE, and the PB RANGE knob beside it, let you set the pitch bend range yourself instead of accepting the range the controller declares. With Override off, the controller's declared range is used. With it on, the PB Range knob sets the range in semitones. This is mainly useful with MPE controllers, which typically declare 48 semitones, when you want something narrower.

PRESSURE SMOOTHER, and the two knobs beside it, smooth incoming aftertouch. MIDI channel pressure is only seven bits, and many controllers send it in visible steps; the smoother turns those steps into a continuous ramp. The two knobs set the rise and fall times independently, so you can have pressure that responds quickly but decays gently.

The remaining two control groups appear only in the standalone iOS version of AD-202. In the plugin formats your host provides the functionality represented by these controls so the panel leaves them out when AD-202 is running as a plugin.

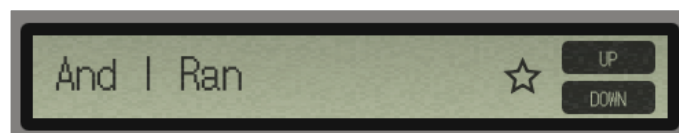
Under LINK, START/STOP SYNC ties the sequencer's RUN button to the Link session's transport in both directions: pressing Run starts every app in the session, and any of them starting will start AD-202. Leave it off if you would rather join the session's tempo without joining its transport. QUANTUM is the bar length, in beats, that Link aligns to when AD-202 joins a session that is already playing; four is a bar of 4/4 and is the usual answer.

The AUDIO / MIDI group is the standalone's device setup: OUTPUT chooses the audio device to play through, SAMPLE RATE sets its rate, MIDI IN chooses the input device to listen to, and the channel setting below it narrows that to a single MIDI channel if you want AD-202 to ignore everything else.

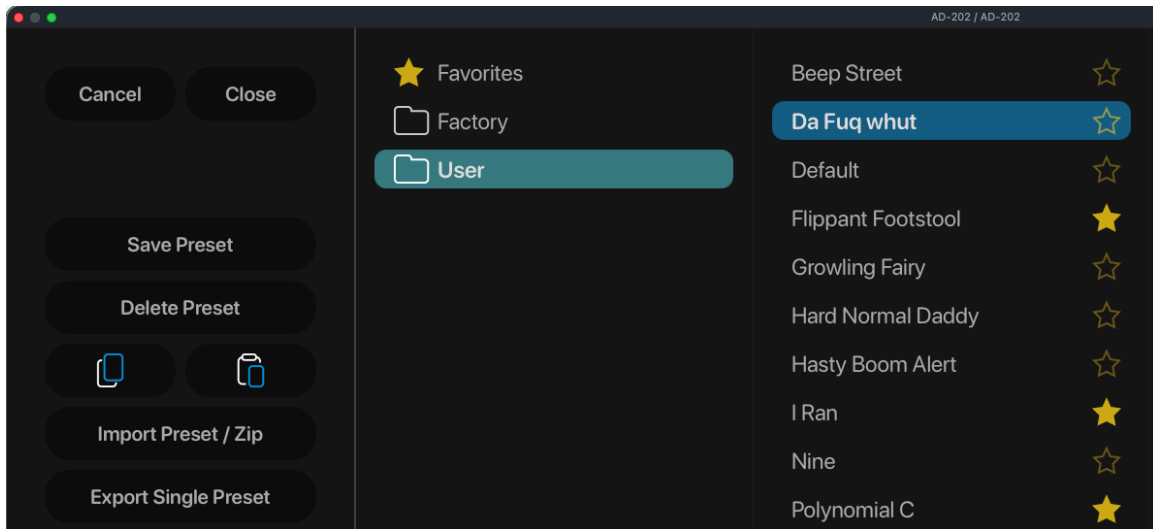


Presets

AD-202 includes a collection of presets to serve as a demonstration of its capabilities and inspirations for your own creations. There are a few controls near the center of the window associated with presets:



The name of the current preset appears in the center. (You probably figured that out yourself.) Clicking the UP and DOWN buttons loads presets in alphabetical order by folder. Clicking the star outline marks the preset as a favorite to help you find it again in the future. To examine all of the presets, click the name of the current to open the preset browser.



The browser displays presets and folders in scrollable lists, arranged in columns. The leftmost list shows the folders within AD-202's preset collection, grouped in two categories: Factory and User. Clicking any of these folders reveals its contents in the next list. Clicking on a preset name loads the settings into AD-202. Click the CLOSE button in the preset browser to dismiss it. If you click the CANCEL button instead, the browser closes and AD-202's settings revert to their previous state. Double-clicking a preset name loads the preset and dismisses the preset browser. Loading a preset irretrievably erases the current settings, so if you have created something that you want to use again, save it as a new preset before loading another preset.

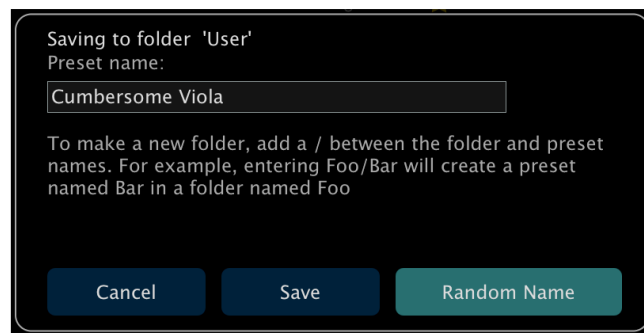
Once you have clicked on any item in the panel, you can navigate within the preset browser with the keys on your keyboard. The left and right arrow keys move the selection between columns, and the up and down arrow keys move it within the list. Tapping the ESC key has the same effect as clicking the Cancel button.

Just like the star at the top of the window which we mentioned previously, the stars to the right of preset names mark presets as favorites. Clicking a filled star removes the mark from a favorite. Once you've marked at least one favorite, a correspondingly named category appears in the leftmost column. Clicking it shows you all the presets you've marked, and clicking their names loads them as usual.

The folders and presets in the browser correspond to folders and files within AD-202's own folder on your storage device (i.e. your computer's hard drive or SSD). This folder is located at `C:\ProgramData\Audio Damage\AD202\` on Windows, and `~/Music/Audio Damage/AD202/` on macOS and Linux. You can store your presets anywhere you like, but for them to show up in the User list they must be placed in the User folder within AD-202's folder. Also, to avoid possible collisions during future updates, do not store your presets within the Factory folder.

Any folders you create within the User folder will show up as folders in the User list. You can create sub-folders within the User folder, and sub-folders within those folders. You can't nest folders deeper than that because the preset browser has only four lists.

To save your presets, click the SAVE PRESET button at the left edge of the window. This invokes a dialog box with a couple of helpful features. As the text therein describes, you can create a folder within the destination folder (whose name is given at the top of the dialog box) by adding the folder's name to the beginning of the preset's name, separated by a slash mark.



Clicking the RANDOM NAME button replaces the preset's name with a pair of words chosen at random from two lists. While the resulting names won't have any connection with what the plugin is doing, you may find this feature useful for coming up with alternatives to routine names like "My Preset 12".

Potential pitfall: once you've saved a preset, clicking its name in the list loads the preset, overwriting whatever changes you've made since you saved the preset. Hence if you want to save the preset again to preserve the changes you've made, do not click on its name before saving it.

You can delete presets and folders from the lists by clicking their name and then clicking the DELETE PRESET or DELETE FOLDER button. AD-202 will give you a chance to confirm this action or cancel it. If you confirm, the preset/folder will be removed from your storage system and is gone for good.

Importing and Exporting Presets

Preset files are plain-text XML files so that you can exchange them online in forums, copy them between a Windows computer and a Macintosh (and even between an iPad and a regular computer), email them to your friends, etc.

The two buttons with icons representing copying and pasting (copy on the left, paste on the right) copy AD-202's current settings to the system clipboard and paste settings from the clipboard. You can use the copy and paste commands to transfer settings between two instances of AD-202 or paste the settings into an email message or text editor. When copied to the clipboard, presets are presented in the same XML text as used in preset files.

The IMPORT PRESET/ZIP button provides a way to add presets to AD-202 without manually moving them into the appropriate folders in your file system. Clicking this button produces a file-browser window wherein you can select either a single preset file or a .zip file containing one or more presets. After you select the file, AD-202 copies the preset(s) into whichever folder you have selected in AD-202's preset list, unzipping the file first if necessary.

Depending on whether you've selected a preset or folder, the EXPORT SINGLE PRESET or EXPORT FOLDER AS ZIP button performs the complementary functions of the Import button. First select either a preset or a folder in AD-202's list, then click the export button. A file-save window appears; choose a location in your file system, give the file a name, and click SAVE. If you have chosen a folder in AD-202's preset list, the plugin places it and all of the presets it contains in a .zip file.

Default Preset

If you save a preset with the special name "Default" in the User folder, new instances of AD-202 will load it automatically when you add it to your DAW session. You can use a default preset file to give yourself the same starting point with AD-202 whenever you use it. The plugin installer creates a default preset file for you but feel free to replace it with your own.

And Finally...

Thanks for purchasing AD-202. We make every effort to ensure your satisfaction with our products and want you to be happy with your purchase. Please write to support@audiodamage.com if you have any questions or comments.

Document Revisions

- 2026-07-28: Initial draft for version 1.0
- 2026-09-24: Updated for version 1.0.5—added the Sequencer chapter and the 1.0.1 and 1.0.5 release notes, corrected the Trigger descriptions, documented the standalone's Link and device settings